

Estonia

Sovereign government data centre network — capacity, legal posture, provider landscape and migration path.

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1 Starting point

Population	1.37 m
GDP	EUR 42 bn
Public administration (NACE O)	37 k
Electricity price	141.0 EUR/MWh
Renewables	38.9%
Live hyperscaler regions	0

2 What is structurally different

- Frontline exposure. A land border with Russia or Belarus (or a Black Sea coast facing the war) changes the threat model from *geopolitical supply disruption* to *kinetic and sabotage risk against the facilities themselves*. Defense and security workloads are scaled up 1.5x/1.25x in the baseline, and at least one site should be hardened (EMP/blast, autonomous power for weeks, not hours). A purely national footprint cannot provide the out-of-country cold copy that Estonia's Data Embassy already demonstrates; this is the first item to revisit when the EU federation layer (Dutch GOAL.md section 16) is modelled.
- No hyperscaler region in-country. Unlike the Netherlands, there is no commercial hyperscale tier to fall back on for non-critical workloads without leaving the jurisdiction. The sovereign core therefore has to be sized for a larger share of total government demand, and the hybrid model (Dutch GOAL.md section 7) needs a cross-border commercial tier.

3 Capacity

Servers	574 (CPU 386 / GPU 30 / storage 158)
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IT critical load	1.0 MW
Facility design load	1.4 MW
Sites	3 (by capacity 1, floor 3)
Average per site	0.5 MW
Site count set by	the minimum-sites floor, not capacity
CAPEX	EUR 35 m
OPEX	EUR 3 m / yr

4 Proposed geography

Region	Role	Share	MW	Notes
Tallinn	Primary civil cloud	41%	0.6	RIT Riigipilv estate, TLLIX, EESF cables to Finland.
Tartu	Sovereign secondary	27%	0.4	University city, 180 km separation; 60 km from the Russian border - hardened design.
Parnu / West	Government / continuity	23%	0.3	Furthest from the eastern frontier; Estonia-Sweden cable.
Out-of-country reserve (Data Embassy)	Strategic reserve	10%	0.1	Existing Luxembourg Data Embassy pattern - national-only model cannot close this gap.

First-pass geographic hypotheses encoding only the obvious constraints, to be replaced by scored site selection.

5 Legal and regulatory posture

Governing instrument	Public Information Act; Data Embassy Act enabling state data held under Estonian jurisdiction abroad
Cloud certification	E-ITS national information security standard (successor to ISKE)
Data classification	State Secrets and Classified Foreign Information Act: Restricted / Confidential / Secret / Top Secret
Procurement route	State Infocommunication Foundation (RIT) as central provider

Foreign jurisdiction exposure. No in-country region; data embassy in Luxembourg is the sovereign offshore pattern

Under the US CLOUD Act and FISA 702 a provider subject to US jurisdiction can face a lawful order for data it holds regardless of where that data sits. Residency is necessary but not sufficient; what matters is who holds the keys and who can be compelled.

6 Current state and provider landscape

Government cloud	Riigipilv (Government Cloud) - RIT (Estonian IT Centre) private state cloud on OCI Dedicated Region + Azure/AWS public-cloud framework (2025); Data Embassy in Luxembourg (2018); X-Road
Maturity	operational
Digital identity	ID-kaart + Mobiil-ID + Smart-ID
Interconnection	TLLIX/RTIX Tallinn; EESF cables to Helsinki; Estlink HVDC

7 Migration path and cost

Phase	Scope	MW	CAPEX	Cumulative	Hybrid
1	Sovereign core	0.3	EUR 5 m	15%	no
2	Security and defense	0.7	EUR 17 m	64%	no
3	State record	0.2	EUR 6 m	80%	no
4	Elective	0.3	EUR 7 m	100%	no

Phase 1 is the number that matters: **EUR 5 m** for 0.3 MW, 15% of total CAPEX. That is the floor below which no hybrid arrangement helps — and it is a small fraction of the full build.

8 Geography and threat notes

No seismic/flood; cold climate and land abundant; FRONTLINE: 300 km Russian border; desynchronised from BRELL Feb 2025 (grid via Estlink/LitPol); repeated subsea sabotage 2023–25; 2007 cyberattack precedent; NATO battlegroup

9 Critical national data in scope

What the platform would *hold*, tiered by consequence of loss rather than by department. Tier 0 is the identity spine; tier 1 is the enforceable relationship between citizen and state. **0 of 15 record classes recorded.**

Tier	Record class	Register	Official description
0	Civil registry core	<i>not yet recorded</i>	
0	Facial biometric	<i>not yet recorded</i>	
0	Fingerprint biometric	<i>not yet recorded</i>	
0	Breeder document scans	<i>not yet recorded</i>	
0	Document issuance history	<i>not yet recorded</i>	
0	Digital identity credentials	<i>not yet recorded</i>	
0	Authentication audit log	<i>not yet recorded</i>	
0	Electoral roll entry	<i>not yet recorded</i>	
1	Tax	<i>not yet recorded</i>	
1	Benefits & pensions	<i>not yet recorded</i>	
1	Land & property registry	<i>not yet recorded</i>	
1	Judicial & criminal justice	<i>not yet recorded</i>	
1	Education	<i>not yet recorded</i>	

Tier	Record class	Register	Official description
1	Business registry	<i>not yet recorded</i>	
1	Vehicle & licensing	<i>not yet recorded</i>	

A blank row means this repository has not yet researched that register. It is not a statement that the country holds no such data. Tiers 2 (health records, imaging) and 3 (genomics, archives, video retention, geospatial) are out of scope: they hold most of the bytes, but Tiers 0 and 1 hold the sovereignty.